

WHYALLA AP, Australia

WMO: 956640

Lat: 33.0539S

Lon: 137.5206E

Elev: 24

StdP: 14.68

Time Zone: 9.50 (AUA)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 35.1	(c) 37.5	(d) 17.3	(e) 13.1	(f) 80.6	(g) 22.5	(h) 17.0	(i) 76.5	(j) 27.7	(k) 63.1	(l) 25.2	(m) 62.5	(n) 5.8	(o) 320	(p) 0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 1	(b) 20.3	(c) 102.9	(d) 66.0	(e) 98.3	(f) 65.3	(g) 93.3	(h) 64.3	(i) 72.7	(j) 83.5	(k) 71.1	(l) 83.0	(m) 69.6	(n) 82.0	(o) 15.7	(p) 350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.2	111.1	75.2	67.9	102.6	75.2	65.9	95.7	74.2	36.4	82.8	35.0	83.3	33.6	81.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 26.3	(b) 23.8	(c) 21.7	(e) 30.6	(f) 111.2	(g) 2.0	(h) 3.2	(i) 29.2	(j) 113.5	(k) 28.0	(l) 115.4	(m) 26.9	(n) 117.2	(o) 25.4	(p) 119.5		
(a) 26.3	(b) 23.8	(c) 21.7	(e) 29.9	(f) 75.8	(g) 2.2	(h) 2.4	(i) 28.3	(j) 77.6	(k) 27.0	(l) 79.0	(m) 25.8	(n) 80.4	(o) 24.2	(p) 82.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	63.5	75.0	74.6	70.3	64.4	57.9	53.2	52.0	53.7	58.5	63.2	68.0	71.4	(6)
(7)		DBStd	10.05	7.01	6.60	6.08	5.54	4.70	4.33	4.25	5.17	6.30	7.05	7.15	6.98	(7)
(8)		HDD50	62	0	0	0	0	1	18	26	16	1	0	0	0	(8)
(9)		HDD65	1826	2	1	13	75	226	356	402	354	220	123	42	12	(9)
(10)		CDD50	4973	776	689	629	432	247	112	89	130	255	410	541	663	(10)
(11)		CDD65	1262	313	271	177	57	7	0	0	3	24	67	133	210	(11)
(12)		CDH74	13601	3130	2621	1646	664	94	3	2	65	419	1058	1621	2278	(12)
(13)		CDH80	6753	1684	1323	769	258	16	0	0	10	145	505	833	1210	(13)
(14)	Wind	WSAvg	10.9	13.7	12.9	11.2	9.4	8.6	8.6	9.3	9.9	10.7	11.9	12.6	12.7	(14)
(15)	Precipitation	PrecAvg	9.8	0.5	0.6	0.7	0.8	1.1	1.1	0.9	1.0	1.3	1.0	0.8	0.7	(15)
(16)		PrecMax	14.3	3.2	2.5	3.1	3.3	2.2	2.8	2.9	2.0	4.2	2.3	1.9	2.6	(16)
(17)		PrecMin	5.8	0.0	0.0	0.0	0.0	0.2	0.2	0.2	0.0	0.0	0.1	0.0	0.0	(17)
(18)		PrecStd	2.4	0.7	0.6	0.8	0.8	0.5	0.7	0.6	0.7	1.1	0.7	0.5	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	109.5	106.2	101.6	94.9	82.2	71.9	72.5	80.8	90.7	97.9	103.6	106.0	(19)
(20)			MCWB	67.5	68.1	64.2	63.5	58.8	56.8	54.1	56.3	59.4	61.3	64.5	66.1	(20)
(21)		2%	DB	103.2	100.2	94.4	87.1	76.1	67.5	68.3	74.7	84.2	91.6	95.9	99.0	(21)
(22)			MCWB	66.7	67.2	64.7	60.9	57.5	54.7	52.8	53.9	57.0	59.5	62.2	64.5	(22)
(23)		5%	DB	96.4	93.6	88.6	81.6	71.8	64.7	65.0	69.8	78.7	85.8	89.8	92.9	(23)
(24)			MCWB	66.5	67.0	64.6	59.7	56.6	53.6	52.0	52.7	55.3	58.1	61.5	63.8	(24)
(25)		10%	DB	88.8	87.9	82.6	76.5	68.2	62.3	62.3	65.3	73.0	79.0	83.1	86.1	(25)
(26)			MCWB	66.1	66.6	64.4	59.8	56.1	53.0	51.4	51.7	54.6	57.1	61.6	63.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.6	77.0	73.2	67.8	63.5	61.1	57.8	59.0	62.7	65.5	69.9	72.2	(27)
(28)			MCD B	85.6	82.7	81.5	74.0	69.4	65.9	64.4	73.5	78.3	80.3	81.7	82.9	(28)
(29)		2%	WB	72.2	73.3	70.7	65.7	61.0	57.7	55.5	56.7	60.0	63.2	67.4	69.6	(29)
(30)			MCD B	84.5	83.6	80.6	73.8	67.9	62.5	62.2	66.5	74.3	76.9	81.8	84.5	(30)
(31)		5%	WB	70.7	71.4	68.7	64.0	59.5	56.1	54.1	55.0	58.3	61.5	65.6	67.7	(31)
(32)			MCD B	83.7	82.7	78.8	73.4	66.7	61.2	61.0	63.6	70.0	74.6	79.7	82.1	(32)
(33)		10%	WB	69.0	69.7	66.8	62.5	58.0	54.6	52.9	53.6	56.7	59.8	63.9	66.1	(33)
(34)			MCD B	82.4	81.6	77.5	72.0	65.5	59.9	59.7	61.9	67.4	73.0	77.2	79.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.3	19.3	19.1	20.2	19.0	17.7	19.9	21.3	22.7	22.2	20.7	20.6	(35)
(36)			MCDBR	35.5	32.2	30.9	31.1	26.1	21.3	24.3	29.1	34.2	36.8	35.8	35.9	(36)
(37)			MCWBR	10.7	10.0	11.6	13.3	12.3	11.4	12.6	13.7	13.5	13.5	11.6	11.1	(37)
(38)		5% WB	MCDBR	23.6	21.0	21.9	21.1	19.2	16.8	19.9	22.7	25.5	26.9	25.6	24.7	(38)
(39)			MCWBR	9.4	8.6	9.9	11.3	11.1	10.9	11.9	12.7	12.1	11.8	9.8	10.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.348	0.348	0.328	0.320	0.308	0.298	0.298	0.308	0.335	0.344	0.350	0.341	(40)	
(41)		taud	2.536	2.562	2.602	2.605	2.616	2.624	2.619	2.573	2.491	2.476	2.497	2.542	(41)	
(42)		Ebn at Noon	314	308	303	289	276	271	277	288	295	305	312	318	(42)	
(43)		Edh at Noon	35	33	30	26	23	22	23	27	32	36	36	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2547	2277	1889	1420	1023	851	927	1233	1672	2078	2361	2527	(44)	
(45)		RadStd	124	118	99	67	41	46	42	68	108	103	104	111	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.41	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		+0.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page